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Transforming Unconventional Well Delivery: Fit-For-Basin Technology Integration and Evolving Drilling Practices for a Threefold Performance Gain

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Abstract

A structured, basin-specific framework was developed for integrating fit-for-purpose technologies with evolving drilling practices to reduce well construction time in unconventional reservoirs. The methodology centers on a Structured Technology Deployment Framework comprising Technology Scouting and Basin-Fit Screening to shortlist solutions aligned with local lithological and geomechanical conditions, a Value–Effort Scoring Framework to prioritize deployments based on quantified operational benefit versus implementation complexity, Controlled Trial and Evaluation with predefined success metrics, and Calculated Risk in Technology Trials to accelerate learning while safeguarding operational integrity.

A structured, cross-functional workflow was implemented to rapidly deploy technologies tailored for unconventional reservoirs. Selection was not limited to a single supplier but based on technical fit-for-basin criteria, including formation properties, pressure regimes, and operational constraints. Technologies were integrated into system-level solutions through close collaboration between drilling, subsurface, fluids, and cementing teams—ensuring alignment across the full well delivery cycle.

Mechanical Earth Models were maintained as live tools, continuously updated with offset data and real-time observations to guide adjustments in BHA design, fluid properties, and casing schemes. The technology refinement cycle, typically spanning several months, was compressed to just a few weeks using structured trial, feedback, and redeployment loops.

The approach emphasized not only ROP but also the mitigation of flat time drivers such as tight-hole sections, wellbore instability, and non-productive events. Optimum operating window envelopes enabled more aggressive execution while staying within geomechanical limits. This integrative, vendor-neutral methodology ensured faster adaptation and consistent performance gains across diverse subsurface conditions.

The integrated, basin-adapted approach led to up to 3 times improvement in ROP, elimination of unnecessary trips, and a fourfold reduction in connection time through continuous circulation. Smarter BHA designs and improved hole cleaning practices prevented tight-hole incidents and reduced instability-related flat time. Real-time model updates enabled proactive adjustments to mud weight and casing depths, enhancing execution efficiency. These combined measures expanded the operational window and improved

overall reliability. The consistent application of fit-for-purpose technologies, aligned with evolving drilling practices and subsurface inputs, enabled faster learning and more repeatable performance. As a result, the well delivery time was reduced up to approximately 70%, with best-in-class wells drilled in one-third of the time compared to the initial campaign baseline.

Unconventional Fields in the United Arab Emirates

Unconventional hydrocarbon resources in the United Arab Emirates (UAE) have become an increasingly important focus of exploration and development, providing a strategic complement to the country's conventional oil and gas reserves. These resources occur primarily in the onshore domains of Abu Dhabi within tight carbonate and organic-rich mudstone successions deposited in intrashelf basins during the Late Cretaceous (Fig. 1). The Shilaif and Diyab formations are the principal targets, recognized for their capacity to generate and retain significant volumes of hydrocarbons (Van Laer et al., 2019; Alharthi et al., 2023).

The unconventional plays extend over large subsurface areas, frequently underlying or in proximity to some of the UAE's most prolific conventional fields. Their broad stratigraphic continuity offers opportunities for long horizontal laterals and pad-based drilling (Raina et al., 2017; Vahrenkamp et al., 2015a). However, subsurface characterization indicates considerable heterogeneity in lithology, organic content, mineralogy, and natural fracture development (Elzeky, 2024; Laer, 2019). This variability occurs both laterally and vertically, influencing reservoir quality, completion efficiency, and ultimate recovery.

From a structural perspective, 3D seismic and regional mapping reveal that the unconventional fairways are segmented by synclines, gentle folds, and fault systems, features that exert strong control on maturity distribution, overpressure development, and localized geomechanical risk (Alharthi, 2022; Makarychev, 2018). Petrophysical evaluation across multiple fields integrates core analysis, NMR, dielectric logs, and mercury injection capillary pressure (MICP) to discriminate moveable hydrocarbons from immobile bitumen and to quantify pore system complexity (Viannet, 2021; Ibe, 2024). Such integrated workflows are essential given that porosity in organic-rich intervals often ranges from 2–5%, increasing only in lean carbonate layers (Van Laer et al., 2019; Raina et al., 2017).

The geomechanical environment adds further complexity: high in-situ stress anisotropy, local zones where minimum horizontal stress approaches overburden, and narrow mud-weight windows are common (Rana, 2024; Steiner, 2015). Overpressured intervals and abrupt mechanical contrasts between organic-rich and clean carbonate benches present drilling hazards such as wellbore instability, tight-hole conditions, and differential sticking risks. These conditions also influence fracture containment during completions and impose constraints on landing depth and trajectory design (Steiner, 2015; Alharthi, 2023).

The combination of heterogeneous reservoir properties, complex structural setting, and challenging geomechanical conditions underscores the need for a development strategy tailored to the UAE unconventional context. While global unconventional experience provides a valuable starting point for tool selection, workflow design, and operational practices, direct transfer of methods from other basins is rarely sufficient. The unique interplay of lithology, stress regime, and operational constraints in UAE unconventional fields requires that such practices and technologies be systematically evaluated, adapted, and optimized for local conditions through structured, data-driven deployment and continuous refinement.

Within the Shilaif Formation, two main reservoir intervals—S1 and S3—are targeted, each subdivided into sub-layers (e.g., Clean, Mix, HT) that exhibit distinct gamma-ray, density, and mechanical signatures. Drilling responses to applied technologies and operating parameters vary markedly between these sub-layers, reinforcing the need for formation-specific BHA designs, parameter envelopes, and technology integration strategies.

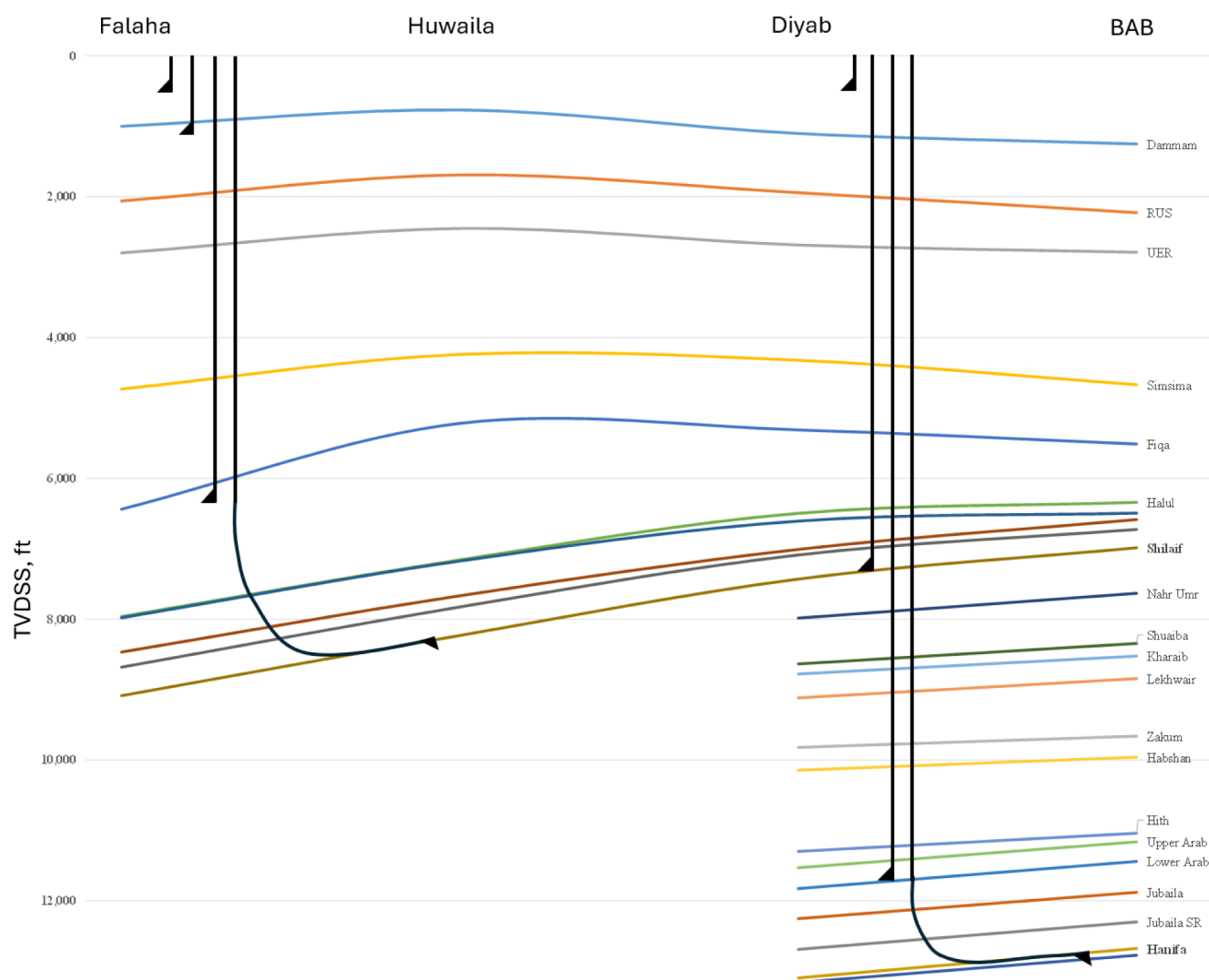


Figure 1—The geological cross-section (simplified view) of the reference unconventional fields.

Structured Technology Deployment Framework

The complexity of UAE unconventional fields necessitates a deliberate and methodical approach to technology introduction. Isolated tool trials or opportunistic adoption cycles often result in inconsistent performance, limited learning, and suboptimal investment of rig time. To address this, a structured framework was established to govern the scouting, evaluation, adaptation, and full-scale application of technologies, ensuring that each deployment was technically justified and basin-specific.

Technology Scouting and Basin-Fit Screening

The first stage involved identifying candidate technologies from both regional and global unconventional developments. Sources included vendor portfolios, operator field trials in analogous basins, and prior regional research (e.g., mechanical earth models, petrophysical integration, drilling performance reviews). Technologies were screened against basin-fit criteria derived from UAE unconventional conditions:

- Compatibility with heterogeneous carbonate–mudstone successions and abrupt mechanical contrasts.
- Tolerance to high stress anisotropy and narrow mud-weight windows.

- Ability to deliver performance improvements without introducing excessive operational complexity.
- Proven adaptability to long-lateral drilling in structurally segmented reservoirs.

Examples of technologies evaluated in this stage include bottom hole assembly (BHA) configurations designed for lateral stability in high dogleg severity (DLS) carbonate intervals, PDC bit designs optimized for interbedded hard/soft lithologies, and shock/vibration mitigation tools with adjustable damping for variable rock strength.

Fig. 2 presents the *effort vs. value* chart, which visualizes the relative complexity and anticipated benefit of implementing specific technologies in unconventional fields in the UAE. This framework is used to prioritize technology deployment by comparing the potential operational value with the practical challenges of implementation.

In this context, the complexity is not limited to the technical difficulty of the application itself. In many cases, the primary driver of complexity is the *absence* of the technology in the country or the lack of established local practices to support it. Consequently, the effort metric reflects both technical and non-technical components. Non-technical factors may include the need to import equipment or materials, establish new supply chains, adapt technology to local environmental or operational conditions, and ensure compliance with regulatory frameworks.

Value (V) Estimation

The value estimation process follows a structured approach:

Scope (S): The total number of wells or projects in which the technology is expected to be deployed.

Impact (I): The per-well benefit of the technology, expressed either in direct monetary savings or in time savings (e.g., reduced rig days). For standardization, any monetary value is converted into *rig days saved*.

Cumulative Impact (C): Calculated as

$$C = I \times S \quad (1)$$

This represents the total potential savings or time reduction across the full scope.

Normalized Value (V): Determined as

$$V = \frac{V_{max}}{C} \quad (2)$$

where V_{max} is the highest cumulative value observed among the evaluated technologies (can be predefined). This normalization allows for consistent cross-comparison.

Effort (E) Estimation Framework

The **Effort** score is derived from a weighted evaluation of multiple dimensions as outlined in the Table 1. Each dimension is assigned a rating and a weight, and the weighted values are summed to obtain the total effort score. The maximum possible score ($E_{max} = 12$) corresponds to the most complex deployment case.

The total Effort Score:

$$E = L + I + G + A_v + C + X, \quad \text{with } 0 \leq E \leq 12 \quad (3)$$

Table 1—Effort scoring framework for technology implementation

Code	Dimension	Score	Rubric (select one level per row)
L	Local Expertise	0–3	0: mature in-country know-how; 1: minor upskilling; 2: vendor led coaching needed; 3: no domestic know-how, full vendor enablement/training required.
I	Integration Complexity	0–2	0: plug-and-play; 1: limited BHA/workflow or data changes; 2: major BHA redesign, digital/EMS re-wiring, multi-team procedure updates.
G	Logistics Complexity	0–2	0: local stock/spares; 1: import with routine lead times; 2: heavy-lift/customs complexity, climate hardening, long lead items.
A _v	Availability & Presence	0–3	0: established in the country, ≥2 approved vendors; 1: available in Area, import to the country with local service base; 2: not in the Area, supplier has regional support only; 3: first Area deployment, no local support.
C	Regulatory Change	0–1	0: within COP; 1: dispensation or COP revision/multi-agency approval required.
X	Exposure (Financial/HSE)	0–1	0: low incremental exposure; 1: high campaign leverage (large scope or added hazard class) even with mitigations.

By plotting value against effort, Fig. 2 provides a visual decision-making tool that highlights "quick wins" (high value, low effort) and identifies technologies that may require significant preparatory work before deployment. This enables stakeholders to strategically allocate resources and plan implementation timelines more effectively.

It is important to note that project value can be enhanced not only through the introduction of new technologies, but also through the re-evaluation and, when necessary, elimination of established practices or previously adopted technologies. In certain cases, technologies that initially appeared promising—often based on successful applications in other regions—may prove to be unsuitable under local conditions.

A clear example from UAE unconventional is the elimination of air drilling. This technique was originally introduced into the unconventional program following positive performance reports from other basins. However, after controlled field trials in the UAE context, the results demonstrated no measurable value contribution. Based on these findings, the decision was made to retire air drilling from the technology portfolio. Importantly, this elimination had no adverse impact on campaign performance and allowed resources - both operational and financial - to be redirected toward higher-value technologies. This highlights the importance of a balanced framework that not only accelerates innovation adoption but also actively filters out low-impact or misfit solutions.

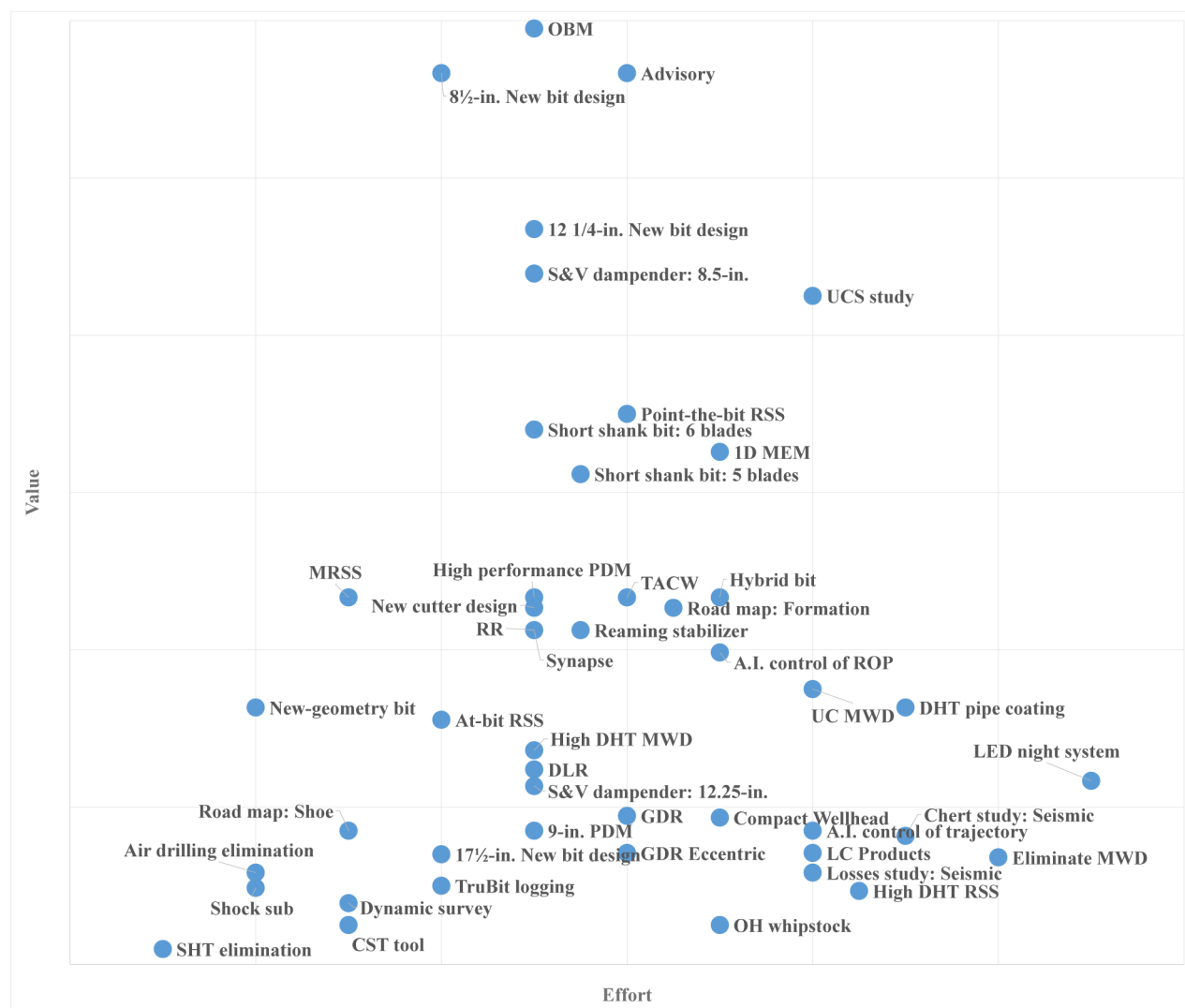


Figure 2—The sample chart shows an example of the effort (complexity) to implement the specific technology vs the value of it for Unconventional operations in Oil fields in UAE.

To further elaborate on the **effort** dimension, the accompanying chart illustrates a roadmap for implementing selected technologies over several quarters. The vertical arrangement of the chart reflects the relative positioning of technology types: solutions related to the *drive mechanism* (e.g., Rotary Steerable Systems) or major BHA change are placed at the top, while other categories—including digital technologies—are positioned in the lower section (Fig. 3).

Technologies that are being introduced in the UAE for the first time are marked with the label "**UAE 1st**", highlighting their novelty in the local operational context. These entries often represent a higher implementation effort due to the absence of prior in-country experience, supply chain readiness, or field-tested best practices.

It is important to note that, although the UAE's conventional fields have been developed for decades, the technological approach required for constructing wells in unconventional fields differs significantly. The challenges in unconventional environments—such as more complex geology, longer laterals, and unique completion designs—demand tailored engineering solutions, modified operational practices, and in many cases, the adoption of advanced or unproven technologies within the UAE context. This difference in baseline conditions is a key driver for the higher effort ratings seen in certain technologies on the chart.



Figure 3—The chart shows the roadmap for the introduction of new technology in unconventional fields over the period of few quarters (Q3, 2024 to Q2, 2025). The first implementation of the technology in UAE marked with "UAE 1st" logo.

Controlled Trial and Evaluation

Based on the develop "technology roadmap" the promising technologies advanced to controlled field trials within selected pilot wells. Each trial was preceded by:

- Defined success metrics (e.g., %ROP improvement, runs per section, vibration signature reduction, connection time improvement).
- Risk assessments addressing wellbore stability, tool survivability, and operational envelope compliance.
- Pre-job readiness reviews with drilling, subsurface, fluids, and completions teams to ensure operational integration.

During execution, real-time monitoring (surface and downhole) enabled immediate parameter optimization. Mechanical earth models were updated live to incorporate drilling response, enabling proactive adjustments for subsequent runs.

Full-Scale Application or Redesign

Post-trial evaluation combined quantitative performance data with operational feedback to determine the next step:

- Full-scale application across the campaign if basin-fit and performance gains were validated.
- Redesign in collaboration with technology providers when adaptation was required for local conditions.
- Retirement of technologies that failed to meet critical performance or reliability thresholds.

Calculated Risk in Technology Trials

A key principle of the roadmap-to-trial framework was the acceptance of calculated risk during controlled field trials. This was not an arbitrary tolerance for failure, but a deliberate, structured approach to decision-making in which the potential benefits of early deployment outweighed the downsides of possible underperformance.

In practice, calculated risk meant that:

- **Failure tolerance was predefined:** Prior to execution, the project team established acceptable performance thresholds and potential failure modes. This ensured that if a trial did not meet all success criteria, the result could still deliver valuable insight without jeopardizing safety or well integrity.
- **Risk exposure was proportional to scope:** Initial deployments were deliberately limited to selected pilot wells or specific sections of the wellbore, containing the operational and financial exposure while still generating representative performance data.
- **Knowledge gain was part of the value proposition:** Even an unsuccessful trial was viewed as a high-value outcome if it revealed critical operational constraints, failure mechanisms, or adaptation requirements for the UAE's unconventional context. These insights often accelerated re-engineering and prevented more costly failures in full-scale campaigns.
- **Proactive mitigation measures were embedded:** Risk assessments informed contingency plans, such as backup tools, alternative drilling parameters, and real-time intervention protocols. This reduced the probability of unplanned downtime or irreversible well impact.

By adopting this mindset, the organization reframed technology trials from a pass/fail exercise into a **learning accelerator**. Instead of viewing underperformance as a setback, it became an input into a rapid improvement loop, allowing design modifications, parameter tuning, or vendor collaboration to occur in parallel with ongoing operations. As a result, the time from initial trial to optimized deployment was reduced from several months to mere weeks, while still protecting operational safety and campaign economics.

Integration into Holistic Well Delivery Systems

Technologies were not deployed in isolation but as part of an integrated solution. For example:

- BHA design optimization was paired with bit design changes to address both stability and ROP.
- Downhole shock/vibration packages were introduced alongside drilling parameter optimization envelopes to maximize tool life.
- Tight-hole mitigation involved both fluid system adjustments and mechanical reaming solutions supported by continuous circulation equipment.
- Well design changes (e.g., casing depth adjustments, profile modifications) were executed to enable the full benefit of new tools.

This integration ensured that the performance of one technology was not undermined by constraints elsewhere in the well delivery process.

Performance Leveraging

Performance improvement in UAE unconventional drilling requires a multi-faceted strategy targeting both drilling speed (ROP optimization) and non-drilling efficiency (flat-time reduction). This is achieved by combining technology upgrades, workflow enhancements, and data-driven parameter management into a single, integrated well delivery system.

Key areas of focus include:

- **ROP Optimization:** bit design, BHA configuration, downhole automation, and drilling parameter envelopes adapted to local rock mechanics.
- **Flat-Time Reduction:** rig automation, remote operations, digital workflows, pre-job readiness, and simultaneous operations.
- **Holistic Integration:** ensuring each technology is deployed as part of a system rather than as an isolated tool.

The following case studies illustrate how this approach was applied in the UAE unconventional context to deliver measurable and repeatable gains.

BHA Design

BHA design in the unconventional wells must reconcile two competing demands:

- Directional responsiveness to achieve geosteering objectives within thin target benches.
- Lateral stability in carbonate intervals with high compressive strength and interbedded softer layers.
- Consistent ROP delivery across variable lithologies.

The specific challenge in reviewed UC Oil fields was related to the well design (Fig. 1), where the 8 1/2-in section is drilled from almost vertical (~ 15 degrees) to horizontal through multiple layers and finished with 6,000 to 10,000 ft lateral length. When such design allows to prevent a major wellbore instability across intermediate 12 1/4-in. section, at the same time it introduces new challenges to 8 1/2-in. section related to major change in downhole temperature (from ~ 150 degF to over 300 degF); increase dogleg severity requirements; severe tight holes during tripping etc.

As stated above, the UC layers have a different response, therefore the different BHA is required for each of them. Depending on the sub-layer the BHA may (or may not) require to deliver higher dogleg severity (DLS); it may require produce higher shock and vibration (as well as stick and slip) while drilling; it may have higher downhole temperature while drilling (exceeding 300 degF) and limiting application of some tools etc. The Fig. 4 shows the high levels schematic approach on various BHA components, depending on specific well requirements.

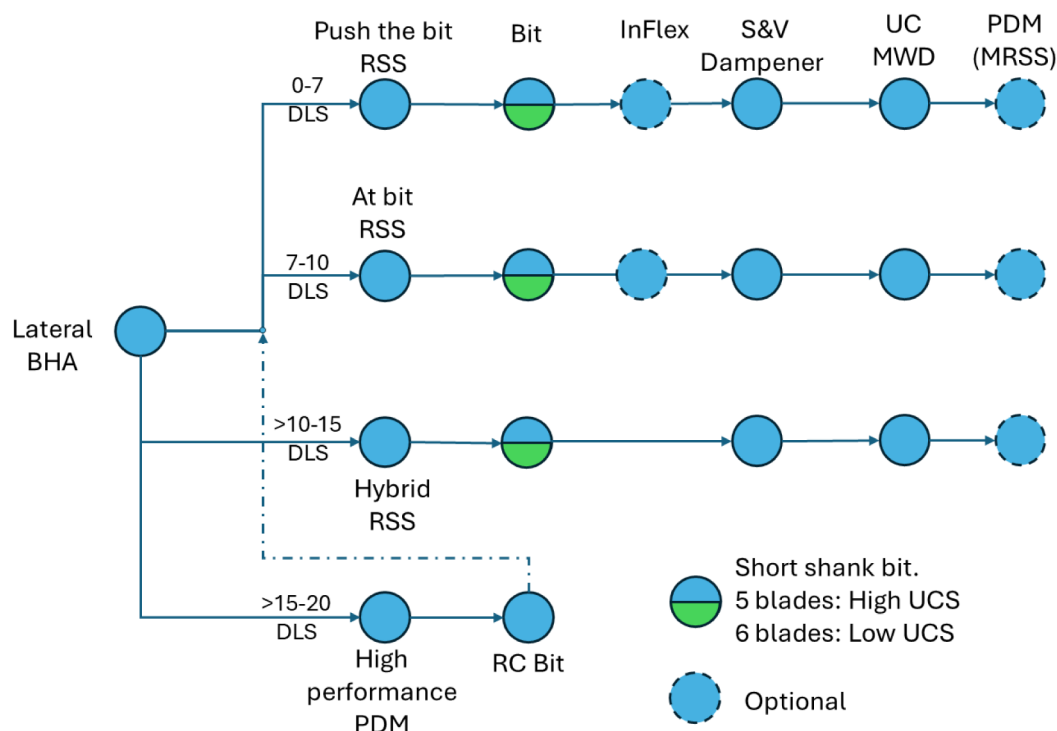


Figure 4—The schematic approach on lateral section BHA selection (DLS unit deg/100ft).

BHA and bit optimization

The bit performance in UAE unconventional drilling is challenged by the high frequency of interbedded layers, which causes cutter impact loading, as well as significant differences in the mechanical properties

of the drilled sub-layers. At the start of the project, the average rate of penetration (ROP) in the field ranged from 44 to 50 ft/hr. This performance required an urgent and structured approach to maximize efficiency in each section.

The overall strategy was shifted from a regressive, reactive model to a progressive, proactive approach. The first step involved a detailed offset well analysis to identify limiters and recurring patterns. Existing technical limits were challenged, and—based on calculated risk—drilling bits were operated at higher thresholds for weight on bit (WOB) and revolutions per minute (RPM). This adjustment alone increased ROP by 30–40% without introducing new technologies.

To achieve the next level of performance, detailed rock characterization was conducted, including measurements of unconfined compressive strength, lithological properties, and other mechanical parameters. All formations were classified, and a preliminary performance roadmap was developed for each interval and formation. The classified rock data was then used to calibrate existing downhole dynamic models. Both the integrated dynamic design and analysis platform and the integrated dynamic system analysis service were employed to simulate bit–BHA interaction and predict optimal operating parameters.

To ensure simulation results were representative, high-frequency dynamic logging was carried out while drilling, closing the data loop. The collected information, combined with post-run dull grading from completed wells, was used to guide either the upgrade of existing bit designs or the initiation of entirely new bit designs ([Table 2](#)).

Table 2—BHA and bits optimization steps

##	Level	Description
1	Offset Well Analysis (drilling data)	Review of historical drilling data from nearby wells to identify limiters, patterns, and baseline performance metrics.
2	Offset Drilling Dynamic Analysis	Post-run analysis of downhole dynamic behaviour (e.g., shock, vibration, torsional oscillations) from offset wells to guide design changes.
3	Offset Wells Rock Characterization	Evaluation of lithology, compressive strength, and mechanical properties from offset wells to understand bit–rock interaction.
4	Rock Library	Centralized database of rock properties and classifications, linked to depth and formation intervals, to support modelling and bit selection.
5	BHA & Rock Calibration	Matching downhole tool configurations with specific rock characteristics to fine-tune predictive models and improve drilling response.
6	Integrated Dynamic Design & Analysis Platform	Software environment combining drilling dynamics modelling, bit/BHA design, and operational parameter simulation in one interface.
7	Integrated Dynamic System Analysis Service	Expert service that interprets dynamic modelling results, validates them against field data, and provides optimization recommendations.
8	Dynamic Memory-Mode Shock & Vibration Logging	High-frequency downhole logging in memory mode to capture detailed dynamic response for validation of models and tool performance.

Implementation of these measures resulted in a threefold increase in ROP across all drilled sections.

Tight holes

Another example of technology, product, and practice introduction aimed at improving performance was the effort to eliminate tight holes in the 12¼-in. section. This section is drilled through the Dammam, Rus, Umm Er Radhuma (UeR), and Simsima formations, with the casing set 600 ft inside the Fiqa formation. The Simsima and UeR formations are composed predominantly of pure limestone, in some intervals with chert

intrusions, and are well known for causing severe tight-hole conditions during pull-out-of-hole (POOH) operations with the drilling BHA. In such cases, back-reaming is required. Once the original BHA is removed, all tight holes disappear, and no wiper trip is required prior to running the casing string.

Tight holes have been observed in unconventional and offset fields for decades, and the problem had not been resolved prior to this project, when up to 30 hours were required to POOH the drilling BHA from section total depth (TD).

To address this challenge, a detailed root-cause investigation was initiated. Based on the findings, a dedicated roadmap was developed, which included:

- Modifications to drilling fluid formulations and practices.
- Trials of alternative drilling practices.
- Changes to BHA design.
- Introduction of new BHA elements aimed at preventing tight-hole formation during POOH operations.

In addition, wireline log data and offset well observations were used to build a one-dimensional mechanical earth model (MEM) for the section.

The implementation of the roadmap yielded positive results. Total back-reaming time was reduced by 75–80%, and in some wells the section was completed without any tight-hole incidents (Fig. 5).

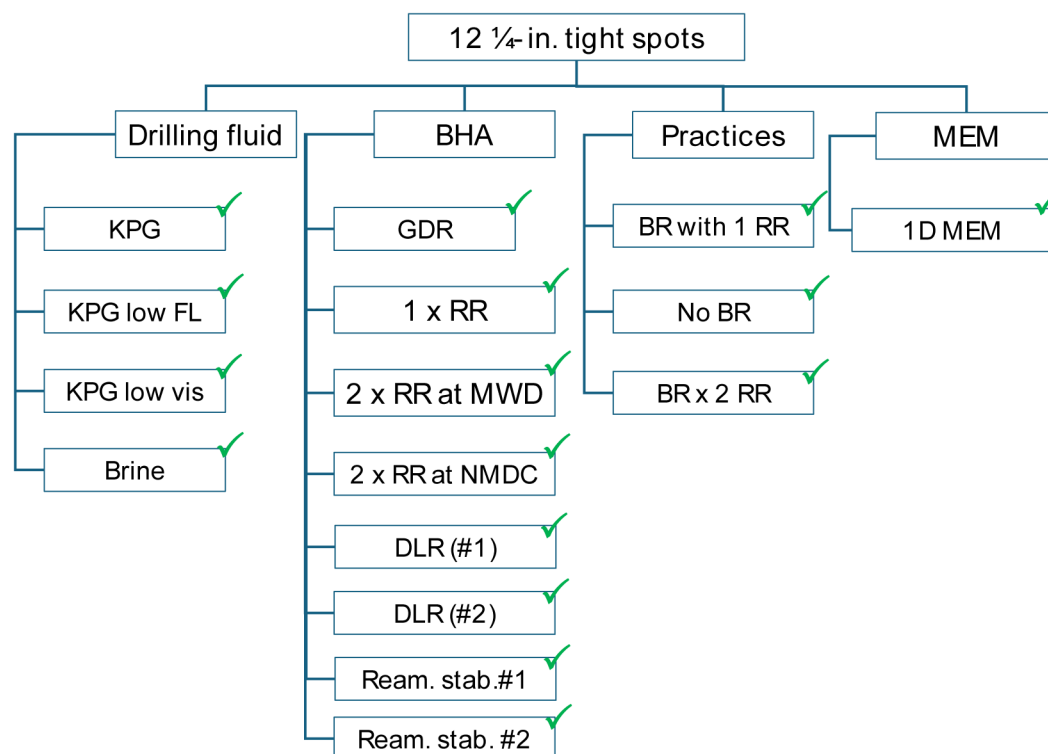


Figure 5—The graph shows the overview of the different trails done to prevent the tight holes across the limestone formations.

Consistency and Repeatability

In the early phase of the unconventional campaign, performance gains were achieved as a result of specific conditions aligning - experienced crews, optimal lithology sequences, or exceptional bit/BHA compatibility. These "record wells" demonstrated the potential for step-change performance, and it was important to ensure the repeatable results across the campaign.

The novel advancement of this program was the deliberate shift from a single-well optimization mindset to a systemic, campaign-wide repeatability model. This was achieved through the convergence of standardized engineering workflows, controlled operational envelopes, and technology integration strategies that maintained performance despite geological and operational variability.

Statistical Evidence of Repeatability

The repeatability means the ability to produce the same result when the same activity is performed multiple times under identical conditions. And in terms of performance the repeatability and improvement refer to ability to achieve the previous positive result and exceed it.

The well construction process can be described as a set of activities which normally repeated in the same order from one well to another. When the downhole conditions (as formation properties) may vary and impact the result, there are activities which improvement is not affected by downhole conditions, as a cased hole tripping speed or a connection time while drilling (Ruzhnikov, 2019). The connection time is the one where collaboration between implemented technologies, quality of the drilling fluid, experience and training of the drilling crew, standardization etc work together. The Fig. 6 is an example of the connection time repeatability and continuous improvement vs time. Total of 6 rigs were operating and even the rigs were onboarded at different time during the project (reflected by local increase in connection time), the overall performance was improving well to well with total improvement close to 4-times.

Similar trend observed in all other key performance indicators (KPI), as tripping speed, ROP etc.

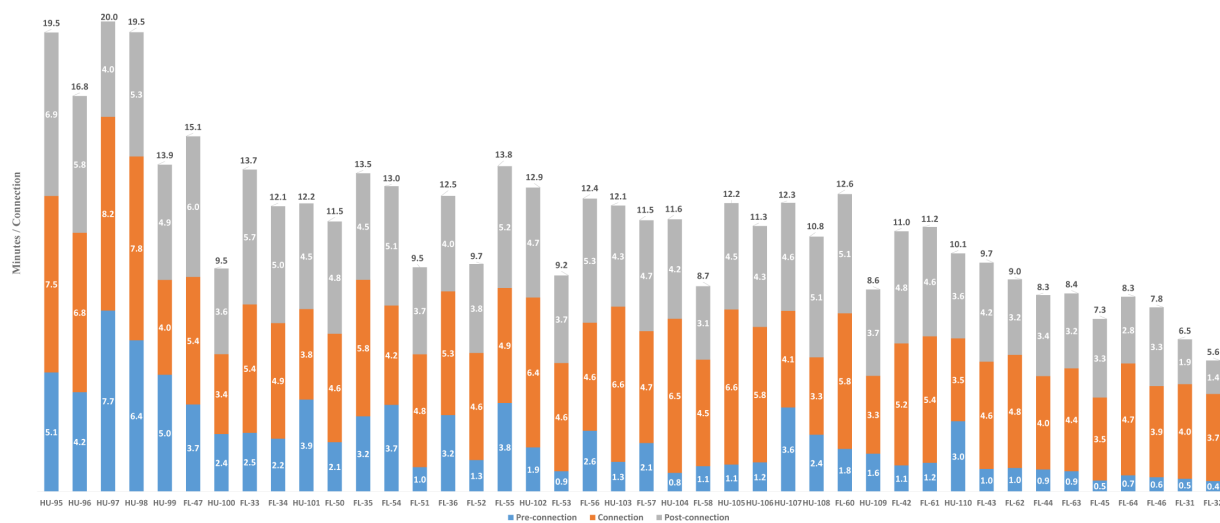


Figure 6—The graph shows the improvement in the connection time over the course of the wells while drilling 12 1/4-in. section.

Operational Envelope Expansion

Achieving repeatable high performance in unconventional drilling is not solely dependent on incremental gains in ROP; it also requires the deliberate expansion of the operational envelope so that optimal practices can be applied across a wider range of lithological and geomechanical conditions. In the UAE unconventional program, this expansion was treated as an engineering objective in its own right, involving both technical modifications and organizational alignment to ensure that risk was managed while capability boundaries were extended.

The process began with early and structured cross-functional engagement. The well placement team was engaged during the planning stage to integrate geosteering objectives with detailed subsurface characterization. This collaboration allowed early identification of intervals with high compressive strength, or severe interbedding, ensuring that BHA and bit designs, as well as drilling parameter envelopes, were pre-tuned to formation-specific challenges. Subsurface, drilling, fluids, and geomechanics teams jointly

validated the expected operating window against MEM predictions and offset well data, avoiding overly conservative constraints while preventing unsafe excursions.

Key technical enablers of the expanded envelope included:

- Higher mechanical loading limits for WOB and RPM in interbedded formations, validated by downhole dynamics modelling and monitored in real time to prevent bit/bearing overload.
- Deployment of high-temperature-rated downhole tools ($>150\text{ }^{\circ}\text{C}$) in deep lateral intervals, enabling sustained high-DLS drilling without tool reliability losses.
- Dynamic BHA selection matrices, linked to the rock property database, which guided assembly configuration changes when transitioning between sub-layers.
- Shorter response time to change the BHA components (i.e. shock dampener), re-design the bit, introduce new technologies.
- Fluid system optimization to maintain ECD stability, hole cleaning, and lubrication in long laterals without requiring wiper trips.
- Optimum Operating Window envelope that allowed the driller to adjust aggressively within safe geomechanical limits without awaiting real-time dispensation.

From an operational perspective, the broadened envelope reduced the gap between "ideal" and "challenging" wells, enabling high-performance drilling in formations previously regarded as limiters. The approach was iterative—each time the envelope was extended in a trial well, results were captured, validated against MEM forecasts, and, if successful, embedded into the campaign's standard operating window.

Conclusions

The proposed approach to technology selection, screening, evaluation, and deployment demonstrates that a structured, data-driven framework can accelerate performance improvement far beyond typical industry learning curves. As illustrated in Fig. 7, a conventional trends for large-scale unconventional operations indicate a gradual, multi-year reduction in well delivery time (orange curve). In contrast, applying an integrated methodology—combining standardized workflows, proactive performance monitoring, and early cross-functional collaboration—can compress this improvement cycle to months (blue curve).

The key features of the approach are integration, the basin-fit adoption of the technologies to ensure technology designs and operational parameters are matched to local lithological and geomechanical conditions, structured approach to technology qualification, with clear decision gates and performance verification stages and continuous improvement through short-cycle feedback loops, operational learning, and dynamic updating of best practices.

A key outcome of this rapid learning cycle was a sustained reduction in well delivery time, with the average spud to rig release duration decreasing by up to **69 %** across the campaign within one year time (Fig. 8). This improvement was achieved without compromising operational safety or well quality, underscoring the effectiveness of a continuous improvement framework embedded in the execution model.

The methodology also demonstrates scalability to other basins, provided that local geological and operational data are integrated into the same structured evaluation framework. This positions the approach not only as a driver of current performance but as a transferable model for future unconventional developments across different geographies and asset types. Future work will focus on enhancing the predictive capability of the methodology through advanced analytics, machine learning integration, and deeper incorporation of real-time drilling dynamics data.

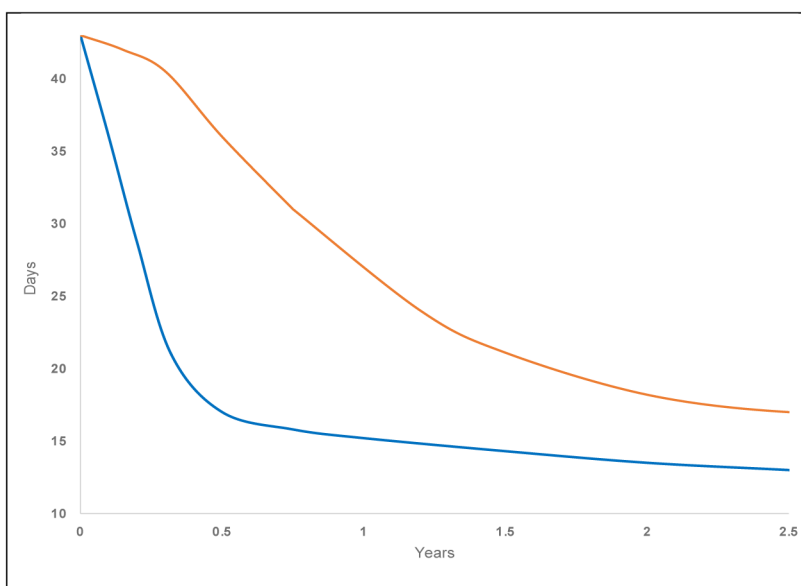


Figure 7—The learning curve

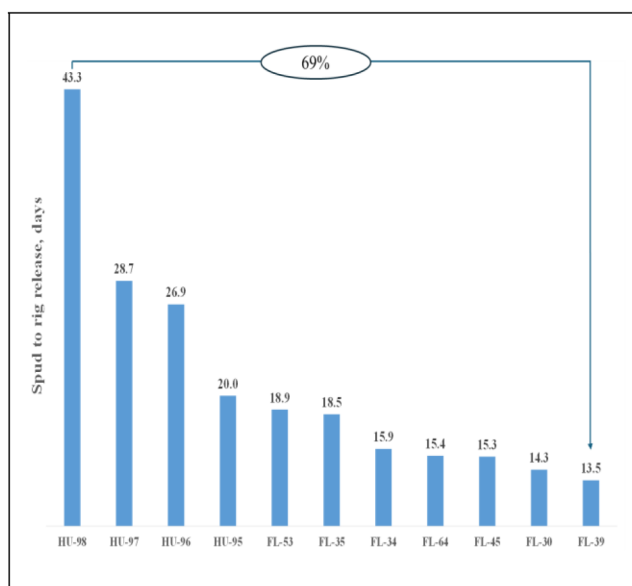


Figure 8—The key well delivery milestones

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